Example 2.57 Find the sum of (i) $1^3+2^3+3^3+\cdots+16^3$ (ii) $9^3+10^3+\cdots+21^3$

Solution (i) $1^3+2^3+3^3+\cdots+16^3=\left[\frac{16\times(16+1)}{2}\right]^2=(136)^2=18496$

(ii) $9^3+10^3+\cdots+21^3=(1^3+2^3+3^3+\cdots+21^3)-(1^3+2^3+3^3+\cdots+8^3)$

$$=\left[\frac{21\times(21+1)}{2}\right]^2-\left[\frac{8\times(8+1)}{2}\right]^2=(231)^2-(36)^2=52065$$

Example 2.58 If $1+2+3+\cdots+n=666$ then find n.

Solution Since, $1+2+3+\ldots+n=\frac{n(n+1)}{2}$, we have $\frac{n(n+1)}{2}=666$

$n^2+n-1332=0 \Rightarrow (n+37)(n-36)=0$

So, $n=-37$ or $n=36$

But $n\neq -37$ ($\because n$ is a natural number); Hence $n=36$.

Progress Check

Say True or False. Justify them.

1. The sum of first n odd natural numbers is always an odd number.
2. The sum of consecutive even numbers is always an even number.
3. The difference between the sum of squares of first n natural numbers and the sum of first n natural numbers is always divisible by 2.
4. The sum of cubes of the first n natural numbers is always a square number.

Exercise 2.9

1. Find the sum of the following series

   (i) $1+2+3+\cdots+60$ (ii) $3+6+9+\cdots+96$ (iii) $51+52+53+\cdots+92$

   (iv) $1+4+9+16+\cdots+225$ (v) $6^2+7^2+8^2+\cdots+21^2$

   (vi) $10^3+11^3+12^3+\cdots+20^3$ (vii) $1+3+5+\cdots+71$

2. If $1+2+3+\cdots+k=325$, then find $1^3+2^3+3^3+\cdots+k^3$.

3. If $1^3+2^3+3^3+\cdots+k^3=44100$ then find $1+2+3+\cdots+k$.

4. How many terms of the series $1^3+2^3+3^3+\cdots$ should be taken to get the sum 14400?

5. The sum of the cubes of the first n natural numbers is 2025, then find the value of n.

6. Rekha has 15 square colour papers of sizes 10 cm, 11 cm, 12 cm,..., 24 cm. How much area can be decorated with these colour papers?

7. Find the sum of the series $(2^3-1^3)+(4^3-3^3)+(6^3-5^3)+\cdots$ to

   (i) n terms (ii) 8 terms

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